

1. Administrative & Study Identification

Full Study Title: A Clinical Study of V540A in Healthy Female Participants (V540A-005) **Short Title/Acronym:** V540A-005 **Protocol Number:** V540A-005 **NCT Number:** NCT07224477 **Sponsor Name:** Merck **Investigational Product:** V540A (Multivalent HPV Vaccine) **Phase of Development:** Phase 2 **Trial Status:** Not Yet Recruiting **Medical Monitor:** Merck Clinical Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate if the investigational vaccine V540A is safe, well-tolerated, and causes an immune response to the 9 shared HPV types that is comparable to the standard vaccine V503 (GARDASIL®9 / SILGARD®9). **Secondary Objectives:** To evaluate the immune response and protective efficacy against additional, unique HPV types not covered by the standard V503 vaccine.

2.2 Background & Rationale Researchers are looking for new, broader ways to prevent cancers related to human papillomavirus (HPV). A standard vaccine (V503) currently protects against 9 types of HPV. V540A is a study vaccine designed to protect against the same 9 shared HPV types plus other unique HPV types, aiming to offer expanded prophylactic protection against HPV-related infections and illnesses in young women.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator-controlled (V503) **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 525 **Number of Sites:** Global and Regional breakdown (Includes sites in the United States, etc.) **Estimated Study Start:** 20-Jul-2026 **Estimated Primary Completion:** 04-Apr-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participant is in good health. 2. Assigned female sex at birth, 16 to 26 years of age. Eligible to participate if not breastfeeding during the study intervention period and for at least 30 days after the last dose of study

intervention. 3. Has a lifetime history of 0 to 4 sexual partners of any sex/gender.

4.2 Exclusion Criteria 1. Has a history of severe allergic reaction (e.g., swelling of the mouth and throat, difficulty breathing, hypotension, or shock) to vaccine components that required medical intervention. 2. Has received a dose of any HPV vaccine before study entry. 3. Has known or history of functional or anatomic asplenia. 4. Is currently immunocompromised or has been diagnosed as having congenital or acquired immunodeficiency, human immunodeficiency virus (HIV) infection, lymphoma, leukemia, systemic lupus erythematosus (SLE), rheumatoid arthritis, juvenile rheumatoid arthritis, inflammatory bowel disease, or other autoimmune conditions. 5. Has thrombocytopenia or other coagulation disorder. 6. Has had a positive test for human papillomavirus (HPV) or has a history of an abnormal cervical biopsy result. 7. Has a history of recurrent respiratory papillomatosis or HPV-related head and neck cancer, HPV-related anal lesions/cancer, HPV-related external genital lesions/cancer, or HPV-related vaginal lesions/cancer. 8. Has received within 12 months before the Day 1 vaccination, is receiving, or plans to receive during the study, immunosuppressive therapies or other therapy known to interfere with the immune response. 9. Is currently receiving systemic steroid therapy or has received 2 or more courses of high-dose corticosteroids lasting at least 1 week in duration in the year prior to Day 1 vaccination. 10. Has received within the 3 months before the Day 1 vaccination, is receiving, or plans to receive during the study any immune globulin product.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Scheduled vaccination doses of V540A (or active comparator V503). **Route of Administration:** Intramuscular (IM) injection. **Duration of Treatment:** Defined vaccination schedule with follow-up extending through the end of the primary completion window (approx. April 2028).

5.2 Concomitant & Prohibited Therapy Permitted: Standard allowed medications for routine, unrelated minor illnesses. **Prohibited:** Immunosuppressive therapies, other investigational vaccines, corticosteroids, immune globulins, or prophylactic treatments that might confound the immunological assessment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Vital Signs, ECG, Pregnancy Test Baseline
Day 0	Randomization, First IM Vaccine Dose, Safety Audit	/	

Diary Dispensation | | **Interim** | Post-Dose | Safety Labs, Immunogenicity/Efficacy Blood Draws, eDiary Review | | **End of Study** | Final Visit | Final Vital Signs, Exit Interview, Antibody Titer Evaluation |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: 1. Number of participants who experience ≥ 1 solicited injection-site Adverse Event (AE) (including redness/erythema, swelling, and pain or tenderness) from Day 1 through Day 5 post each vaccination. 2. Number of participants who experience ≥ 1 solicited Pyrexia AE (defined as a temperature $\geq 100.4^{\circ}\text{F}$ / $\geq 38.0^{\circ}\text{C}$) from Day 1 through Day 5 post each vaccination. 3. Number of participants who experience ≥ 1 Vaccine-related Serious AE (SAE). **Measurement Method:** Patient-reported outcomes via eDiary, laboratory analysis, and standardized investigator AE tracking.

7.2 Secondary Endpoints 1. Geometric Mean Titers (GMTs) of each of the shared and unique HPV vaccine types at Day 29 Postdose 1. 2. Geometric Mean Titers (GMTs) of each of the shared and unique HPV vaccine types at Day 29 Postdose 2. 3. Geometric Mean Titers (GMTs) to each of the shared and unique HPV vaccine types at Day 29 Postdose 3.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient diary (eDiary) for local injection-site reactions (Days 1-5), fever tracking (Days 1-5), and HCP interviews during follow-up visits. **Serious Adverse Events (SAEs):** Monitored throughout the study and reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent safety monitoring mechanisms managed by the sponsor to oversee trial safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary safety outcomes; Per-Protocol (PP) for immunogenicity endpoints. **Sample Size Justification:** Powered to demonstrate immunological non-inferiority of V540A vs V503 for the shared HPV serotypes ($\alpha = 0.05$). **Handling of Missing Data:** Standard missing data imputations for immunogenicity dropouts as specified by the SAP.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits to ensure protocol compliance and correct IM administration. **Audit Trail:** Complete eCRF locking, tracking of electronic patient diaries, and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: V540A-005 **Registry Number:** NCT07224477 **Sponsor/Lead Organization:** Merck **Study Title:** A Clinical Study of V540A in Healthy Female Participants (V540A-005) **Official Regulatory Title:** A Phase 2, Randomized, Double-Blind, Comparator-Controlled Study to Evaluate the Safety, Tolerability, and Immunogenicity of V540A in Healthy Females 16 to 26 Years of Age

12. Clinical Framework (HPV Prevention Specific)

Primary Condition: Human Papillomavirus (HPV) Infections / HPV-related cancers **Diagnosis Threshold:** Healthy participants (Prophylactic targeted population) **Medical Methodology:** Prophylactic Immunization **Optimization Target:** Elicitation of an active immune response to multivalent HPV types

13. Study Procedures & Methodology

Management Pathway: Standard vaccine administration pathway **Education Protocol (HCP):** Protocol-specific IM administration, blinding maintenance, and AE reporting training **Education Protocol (Patient):** Onsite training for utilizing patient eDiaries to log post-vaccination symptoms (erythema, swelling, pain, pyrexia) **Quality Audit Mechanism:** HCP protocol adherence tracking and patient eDiary compliance monitoring

14. Observation & Follow-Up Schedule

Total Duration: Follow-up extending until study completion in early 2028 **Onsite Visit Cadence:** Scheduled tightly around vaccination doses and specific immunogenicity blood-draw timepoints (e.g., Day 29 post each dose) **Remote Monitoring Frequency:** Routine post-vaccination safety follow-ups **Checklist Review Interval:** eDiary and AE reviews at each designated study interval

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |